

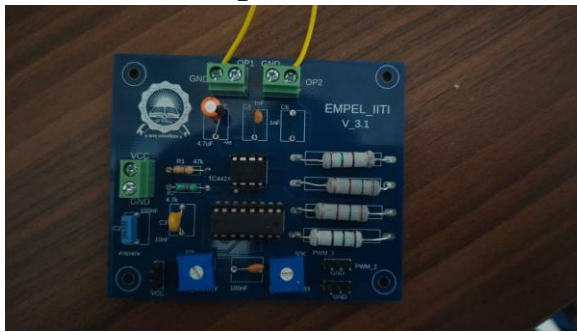
EE 252 LAB BUCK CONVERTER PROJECT REPORT (G1-B2)

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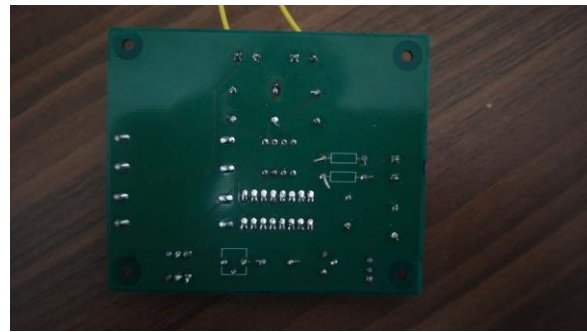
Specifications

- **Parameters:** Input 12.5 V, Output 7.5 V, Switching frequency 12.5 kHz, Output current 1 A.

Hardware Implementation



(a) Gate Driver Board



(b) Soldered Driver

Figure 1: Hardware Assembly Components



(a) Complete Setup



(b) Output Voltage

Figure 2: Experimental Setup and Measured Output

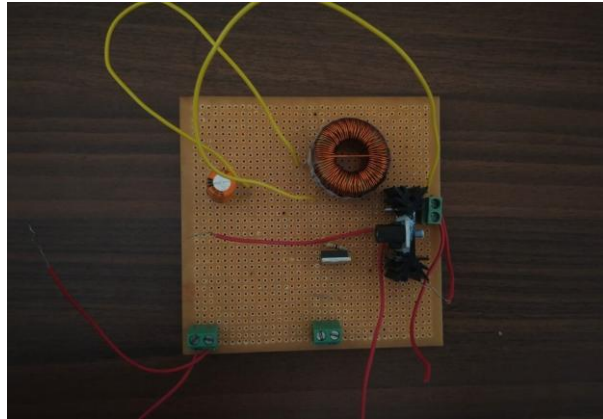


Figure 3: Perf Board

Waveform Analysis

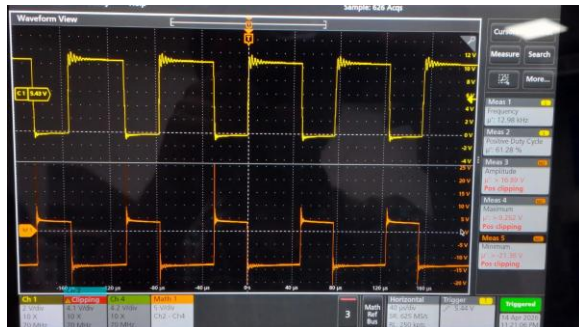


(a) Inductor Current (CCM)



(b) Inductor Current (DCM)

Figure 4: Comparison of Inductor Currents



(a) Diode Voltage



(b) Switch Current

Figure 5: Hardware Waveforms

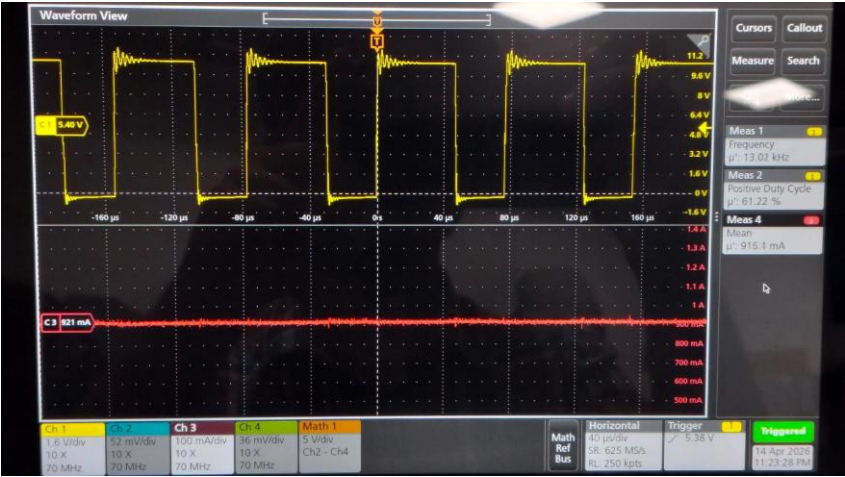
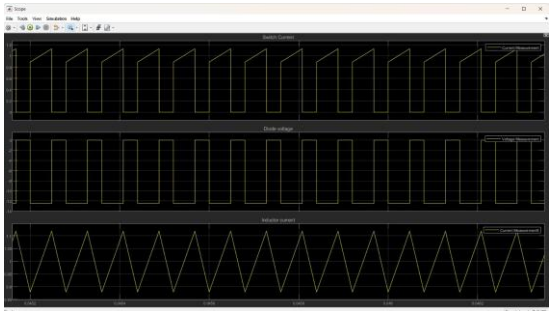


Figure 6: Output Current

MATLAB Simulation



(a) CCM Waveform



(b) DCM Waveform (1kHz)

Figure 7: MATLAB Simulation Waveform

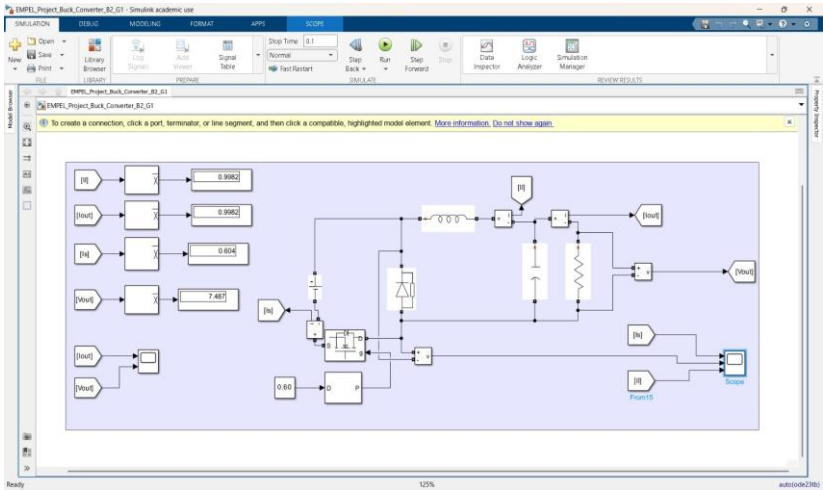


Figure 8: Buck converter circuit